Academic Programme

Curriculum Structure & Syllabus

BS Degree in Data Science and Applications

Become an IITian

Build Your Future, Power India's AI Revolution

Foundation	Diploma	BSc Degree	BS Degree
32 Credits | 8 Courses	54 Credits | 2 Streams	114 Credits Total	142 Credits Total

.

Programme Architecture

Four progressive academic levels designed from fundamentals to advanced applications

1. Foundation	2. Diploma	3. BSc Degree	4. BS Degree
Maths, Statistics, English, Computational Thinking, Python	Programming + Data Science streams with projects	Advanced credits leading to BSc award	Final advanced credits leading to BS award

Credit Pathway

Academic Milestone	Credits at Level	Award/Progression
Foundation Level	32	Eligibility to progress to Diploma Level
Programming Diploma	27	Diploma in Programming
Data Science Diploma	27	Diploma in Data Science
BSc Degree Level	28	BSc Degree: 114 total credits
BS Degree Level	28	BS Degree: 142 total credits

Contents

1 Foundation Level

Foundation Level (Including Pre-Qualifier) Curriculum Structure

Program Link: `https://study.iitm.ac.in/ds/`

Total Credits: 32

1. Course Structure

S. No.	Course Name	Credits	Code	Pre-requisites	Co-requisites
1	Mathematics for Data Science I	4	BSMA1001	None	None
2	Statistics for Data Science I	4	BSMA1002	None	None
3	Computational Thinking	4	BSCS1001	None	None
4	English I	4	BSHS1001	None	None

5	Mathematics for Data Science II	4	BSMA1003	BSMA1001	None
6	Statistics for Data Science II	4	BSMA1004	BSMA1001, BSMA1002	BSMA1003
7	Programming in Python	4	BSCS1002	BSCS1001	None
8	English II	4	BSHS1002	BSHS1001	None

Total Credit Hours: 32

2. Qualifier Process (Regular Entry)

All regular entry applicants must successfully complete a Qualifier Process to gain admission into the Foundation Level.

2.1 Qualifier Preparation (4 Weeks)

Duration: 4 weeks

Courses covered: English I, Mathematics for Data Science I, Statistics for Data Science I, Computational Thinking

2.2 Qualifier Exam

A Qualifier Examination is conducted after 4 weeks based on the studied content.

3. Eligibility Criteria for Qualifier Exam

Eligibility is based on assignment performance.

Average of best 2 out of first 3 assignments per course is considered.

Minimum Average Assignment Score

Category	Minimum Score
General Learner	40%
SC / ST / PwD (40% disability)	30%
PwD (40% disability) & SC / ST	30%
OBC-NCL / EWS	35%

4. Important Note

Relaxations apply only for the Qualifier Process.

No relaxations apply after admission into the program.

Detailed Syllabus of Courses:

Foundational Level Courses (Sr. No. 1 -8):

1. Mathematics for Data Science I

This course introduces functions (straight lines, polynomials, exponentials and logarithms) and discrete mathematics (basics, graphs) with many examples. The students will be exposed to the idea of using abstract mathematical structures to represent concrete real-life situations.

Course ID: BSMA1001

Course Credits: 4

Course Type: Foundational

Pre-requisites: None

12 weeks of coursework, weekly assignments, 2 in-person invigilated quizzes, 1 in-person invigilated end term exam. For details of standard course structure and assessments, visit Academics page.

WEEK 1 Set Theory - Number system, Sets and their operations, Relations and functions - Relations and their types, Functions and their types

WEEK 2 Rectangular coordinate system, Straight Lines - Slope of a line, Parallel and perpendicular lines, Representations of a Line, General equations of a line, Straight-line fit

WEEK 3 Quadratic Functions - Quadratic functions, Minima, maxima, vertex, and slope, Quadratic Equations

WEEK 4 Algebra of Polynomials - Addition, subtraction, multiplication, and division, Algorithms, Graphs of Polynomials - X-intercepts, multiplicities, end behavior, and turning points, Graphing & polynomial creation

WEEK 5 Functions - Horizontal and vertical line tests, Exponential functions, Composite functions, Inverse functions

WEEK 6 Logarithmic Functions - Properties, Graphs, Exponential equations, Logarithmic equations

WEEK 7 Sequence and Limits - Function of One variable • Function of one variable • Graphs and Tangents • Limits for sequences • Limits for function of one variable • Limits and Continuity

WEEK 8 Derivatives, Tangents and Critical points - • Differentiability and the derivative • Computing derivatives and L'Hopital's rule • Derivatives, tangents and linear approximation • Critical points: local maxima and minima

WEEK 9 Integral of a function of one variable - • Computing areas, Computing areas under a curve, The integral of a function of one variable • Derivatives and integrals for functions of one variable

WEEK 10 Graph Theory - Representation of graphs, Breadth-first search, Depth-first search, Applications of BFS and DFS; Directed Acyclic Graphs - Complexity of BFS and DFS, Topological sorting

WEEK 11 Longest path, Transitive closure, Matrix multiplication

Graph theory Algorithms - Single-source shortest paths, Dijkstra's algorithm, Bellman-Ford algorithm,

All-pairs shortest paths, Floyd–Warshall algorithm,

Minimum cost spanning trees, Prim's algorithm, Kruskal's algorithm

WEEK 12 Revision

2. Statistics for Data Science I

The students will be introduced to large datasets. Using this data, the students will be introduced to various insights one can glean from the data. Basic concepts of probability also will be introduced during the course leading to a discussion on Random variables.

Course ID: BSMA1002

Course Credits: 4

Course Type: Foundational

Pre-requisites: None

12 weeks of coursework, weekly assignments, 2 in-person invigilated quizzes, 1 in-person invigilated end term exam. For details of standard course structure and assessments,

WEEK 1 Introduction and type of data, Types of data, Descriptive and Inferential statistics, Scales of measurement

WEEK 2 Describing categorical data Frequency distribution of categorical data, Best practices for graphing categorical data, Mode and median for categorical variable

WEEK 3 Describing numerical data Frequency tables for numerical data, Measures of central tendency - Mean, median and mode, Quartiles and percentiles, Measures of dispersion - Range, variance, standard deviation and IQR, Five number summary

WEEK 4 Association between two variables - Association between two categorical variables - Using relative frequencies in contingency tables, Association between two numerical variables - Scatterplot, covariance, Pearson correlation coefficient, Point bi-serial correlation coefficient

WEEK 5 Basic principles of counting and factorial concepts - Addition rule of counting, Multiplication rule of counting, Factorials

WEEK 6 Permutations and combinations

WEEK 7 Probability Basic definitions of probability, Events, Properties of probability

WEEK 8 Conditional probability - Multiplication rule, Independence, Law of total probability, Bayes' theorem

WEEK 9 Random Variables - Random experiment, sample space and random variable, Discrete and continuous random variable, Probability mass function, Cumulative density function

WEEK 10 Expectation and Variance - Expectation of a discrete random variable, Variance and standard deviation of a discrete random variable

WEEK 11 Binomial and poisson random variables - Bernoulli trials, Independent and identically distributed random variable, Binomial random variable, Expectation and variance of abinomial random variable, Poisson distribution

WEEK 12 Introduction to continous random variables - Area under the curve, Properties of pdf, Uniform distribution, Exponential distribution

Prescribed Books

The following are the suggested books for the course:

Introductory Statistics (10th Edition) - ISBN 9780321989178, by Neil A. Weiss published by Pearson

Introductory Statistics (4th Edition) - by Sheldon M. Ross

3. Computational Thinking

The students will be introduced to a number of programming concepts using illustrative examples which will be solved almost entirely manually. The manual execution of each solution allows for close inspection of the concepts being discussed.

Course ID: BSCS1001

Course Credits: 4

Course Type: Foundational

Pre-requisites: None

12 weeks of coursework, weekly online assignments, 2 in-person invigilated quizzes, 1 in-person invigilated end term exam. For details of standard course structure and assessments, visit Academics page.

WEEK 1 Variables, Initialization, Iterators, Filtering, Datatypes, Flowcharts, Sanity of data

WEEK 2 Iteration, Filtering, Selection, Pseudocode, Finding max and min, AND operator

WEEK 3 Multiple iterations (non-nested), Three prizes problem, Procedures, Parameters, Side effects, OR operator

WEEK 4 Nested iterations, Birthday paradox, Binning

WEEK 5 List, Insertion sort

WEEK 6 Table, Dictionary

WEEK 7 Graph, Matrix

WEEK 8 Adjacency matrix, Edge labelled graph

WEEK 9 Backtracking, Tree, Depth First Search (DFS), Recursion

WEEK 10 Object oriented programming, Class, Object, Encapsulation, Abstraction, Information hiding, Access specifiers

WEEK 11 Message passing, Remote Procedure Call (RPC), Cache memory, Parallelism, Concurrency, Polling, Preemption, Multithreading, Producer Consumer, Atomicity, Consistency, Race condition, Deadlock, Broadcasting

WEEK 12 Top-down approach, Bottom-up approach, Decision tree, Numerical prediction, Behaviour analysis, Classification

4. English I

This course aims at achieving fluency and confidence in spoken and written English. This course will use insights from theories of learning and dominant methods of teaching language.

Course ID: BSHS1001

Course Credits: 4

Course Type: Foundational

Pre-requisites: None

12 weeks of coursework, weekly online assignments, 2 in-person invigilated quizzes, 1 in-person invigilated end term exam. For details of standard course structure and assessments.

WEEK 1	Sounds and Words (Vowel and Consonant sounds)
WEEK 2	Parts of Speech
WEEK 3	Sentences (Phrases and Idioms)
WEEK 4	Speaking Skills (Spoken English Preliminaries)
WEEK 5	Tenses and Agreement in English Sentences
WEEK 6	Reading Skills (Skimming, Scanning and Comprehension)
WEEK 7	Listening Skills
WEEK 8	Aspiration, Word Stress and Syllabification
WEEK 9	Speaking Skills (Presentation and Group Discussion)
WEEK 10	Grammar (Common Errors in English) and Writing Skills
WEEK 11	Writing Skills (Basics of Writing)
WEEK 12	Writing Skills (Professional Writing)

Prescribed Books

The following are the suggested books for the course:

Aarts, Bas (2011). Oxford Modern English Grammar, New York: Oxford University Press

Murphy, Raymond (2012). English Grammar in Use, New York: Cambridge University Press. 4th Edition

Krishnaswamy, Subashree and K. Srilata eds. (2007). Short Fiction from South India. Delhi: OUP.

Dhanavel, S.P. (2010). English and soft skills (V-1). Chennai: Orient Blackswan.

5. Mathematics for Data Science II

This course aims to introduce the basic concepts of linear algebra, calculus and optimization with a focus towards the application area of machine learning and data science.

Course ID: BSMA1003

Course Credits: 4

Course Type: Foundational

Pre-requisites: BSMA1001 - Mathematics for Data Science I

11 weeks of coursework, weekly online assignments, 2 in-person invigilated quizzes, 1 in-person invigilated end term exam. For details of standard course structure and assessments, visit Academics page.

WEEK 1 Vector and matrices - Vectors; Matrices; Systems of Linear Equations; Determinants (part 1); Determinants (part 2)

WEEK 2 Solving linear equations - Determinants (part 3); Cramer's Rule; "Solutions to a system of linear equations with an invertible coefficient matrix"; The echelon form; Row reduction; The Gaussian elimination method

WEEK 3 Introduction to vector spaces - Introduction to vector spaces; Some properties of vector spaces; Linear dependence; Linear independence - Part; Linear independence - Part 2

WEEK 4 Basis and dimension - What is a basis for a vector space?; Finding bases for vector spaces; What is the rank/dimension for a vector space; Rank and dimension using Gaussian elimination

WEEK 5 Rank and Nullity of a matrix; Introduction to Linear transformation - The null space of a matrix: finding nullity and a basis - Part 1; The null space of a matrix: finding nullity and a basis - Part 2; What is a linear mapping - Part 1; What is a linear mapping - Part 2; What is a linear transformation

WEEK 6 Linear transformation, Kernel and Images - Linear transformations, ordered bases and matrices; Image and kernel of linear transformations; Examples of finding bases for the kernel and image of a linear transformation

WEEK 7 Equivalent and Similar matrices; Introduction to inner products - Equivalence and similarity of matrices; Affine subspaces and affine mappings; Lengths and angles; Inner products and norms on a vector space

WEEK 8 Orthogonality, Orthonormality; Gram-schmidt method - Orthgonality and linear independence; What is an orthonormal basis? Projections using inner products; The Gram-Schmidt process; Orthogonal transformations and rotations

WEEK 9 Multivariable functions, Partial derivatives, Limit, continuity and directional derivatives - Multivariable functions: visualization; Partial derivatives; Directional derivatives; Limits for scalar-valued multivariable functions; Continuity for multivariable functions; Directional derivatives in terms of the gradient

WEEK 10 Directional ascent and descent, Tangent (hyper) plane, Critical points - The directional of steepest ascent/descent; Tangents for scalar-valued multivariable functions; Finding the tangent hyper(plane); Critical points for multivariable functions

WEEK 11 Higher order partial derivatives, Hessian Matrix and local extrema, Differentiability - Higher order partial derivatives and the Hessian matrix; The Hessian matrix and local extrema for f(x,y); The Hessian matrix and local extrema for f(x,y,z); Differentiability for Multivariable Functions; Review of Maths - 2

6. Statistics for Data Science II

This second course will develop on the first course on statistics and further delve into the main statistical problems and solution approaches

Course ID: BSMA1004

Course Credits: 4

Course Type: Foundational

Pre-requisites: BSMA1002 - Statistics for Data Science I

BSMA1001 - Mathematics for Data Science I

Co-requisites: BSMA1003 - Mathematics for Data Science II

12 weeks of coursework, weekly online assignments, 2 in-person invigilated quizzes, 1 in-person invigilated end term exam. For details of standard course structure and assessments, visit Academics page.

WEEK 1 Multiple random variables - Two random variables, Multiple random variables and distributions

WEEK 2 Multiple random variables - Independence, Functions of random variables - Visualization, functions of multiple random variables

WEEK 3 Expectations Casino math, Expected value of a random variable, Scatter plots and spread, Variance and standard deviation, Covariance and correlation, Inequalities

WEEK 4 Continuous random variables Discrete vs continuous, Weight data, Density functions, Expectations

WEEK 5 Multiple continuous random variables - Height and weight data, Two continuous random variables, Averages of random variables - Colab illustration, Limit theorems, IPL data - histograms and approximate distributions, Jointly Gaussian random variables Probability models for data - Simple models, Models based on other distributions, Models with multiple random variables, dependency, Models for IPL powerplay, Models from data

WEEK 6 Refresher week

WEEK 7 Estimation and Inference I

WEEK 8 Estimation and Inference II

WEEK 9 Bayesian estimation

WEEK 10 Hypothesis testing I

WEEK 11 Hypothesis Testing II

WEEK 12 Revision week

Prescribed Books

The following are the suggested books for the course:

Probability and Statistics with Examples using R.

Author: Siva Athreya, Deepayan Sarkar and Steve Tanner

7. Programming in Python

This will be the first formal programming course that students will see in this programme. The goal of this course is to introduce Python programming, which is used throughout the programme, with a basic problem solving and algorithmic flavour.

Course ID: BSCS1002

Course Credits: 4

Course Type: Foundational

Pre-requisites: BSCS1001 - Computational Thinking

12 weeks of coursework, weekly online assignments, 2 in-person invigilated quizzes, 1 in-person invigilated end term exam. For details of standard course structure and assessments, visit Academics page.

WEEK 1	**Introduction to algorithms**
WEEK 2	Conditionals
WEEK 3	Conditionals (Continued)
WEEK 4	Iterations and Ranges
WEEK 5	Iterations and Ranges (Continued)
WEEK 6	Basic Collections in Python
WEEK 7	Basic Collections in Python (Continued)
WEEK 8	Basic Collections in Python (Continued)
WEEK 9	File Operations
WEEK 10	File Operations (Continued)
WEEK 11	Module system in python
WEEK 12	Basic Pandas and Numpy processing of data

Prescribed Books

The following are the suggested books for the course:

Title: Python for Everybody.

Author: Charles R. Severance.

Publisher: Shroff Publishers.

ISBN: 9789352136278

(The PDF of this book is currently available freely at

http://do1.dr-chuck.com/pythonlearn/EN_us/pythonlearn.pdf)

8. English II

Focus on achieving greater degree of fluency in functional and conversational English to understand subtle and detailed meaning in conversations and texts through short literary pieces and contextualized content.

Course ID: BSHS1002

Course Credits: 4

Course Type: Foundational

Pre-requisites: BSHS1001 - English I

12 weeks of coursework, weekly online assignments, 2 in-person invigilated quizzes, 1 in-person invigilated end term exam. For details of standard course structure and assessments.

WEEK 1	Patterns in Sentences
WEEK 2	Patterns in Sentences (Continued)
WEEK 3	Patterns in Sentences (Continued)
WEEK 4	Listening Skills
WEEK 5	Listening Skills (Continued)
WEEK 6	Speaking Skills
WEEK 7	Speaking Skills (Continued)
WEEK 8	Reading Skills
WEEK 9	Writing Skills
WEEK 10	Writing Skills (Continued)
WEEK 11	Social Skills
WEEK 12	Social Skills (Continued)

2 Diploma Level

Diploma Level Curriculum Structure

Diploma Level consists of two streams:

Diploma in Programming and Diploma in Data Science.

Each diploma includes 5 core courses, 2 projects, and 1 skill enhancement course.

Requirements for Registration

Learner must successfully clear all 8 Foundation Level courses.

Exit / Progression Options

Learners may progress or exit based on completion:

- Proceed to BSc Degree Level after completing Foundation + both Diplomas
- Exit with Diploma in Programming (IIT Madras)

- Exit with Diploma in Data Science (IIT Madras)
- Exit with both Diplomas (IIT Madras)

Diploma in Programming

Focus: Databases, programming, data structures, algorithms, and web application development.

Duration: 1–2 years | Workload: 15 hrs/week | Credits: 27 (6 courses + 2 projects)

Course Structure - Diploma in Programming

Course Name	Credits	Code	Prerequisites	Corequisites
Database Management Systems	4	BSCS2001	None	None
Programming, Data Structures and Algorithms using Python	4	BSCS2002	None	None
Modern Application Development I	4	BSCS2003	None	BSCS2001
PROJECT Modern Application Development I	2	BSCS2003P	None	BSCS2003
Programming Concepts using Java	4	BSCS2005	None	None

Modern Application Development II	4	BSCS2006	BSCS2003	None
PROJECT Modern Application Development II	2	BSCS2006P	BSCS2003P	BSCS2006
System Commands	3	BSSE2001	None	None

Diploma in Data Science

Focus: Data analysis, machine learning, business data interpretation, and AI applications.

Credits: 27 | Duration: 1–2 years | Workload: 15 hrs/week

Students complete 21 mandatory credits (5 courses + 1 project) and choose one of two tracks:

Option 1: Business Analytics + Project

Option 2: Deep Learning & Generative AI + Project

Course Structure - Diploma in Data Science

Course Name	Credits	Code	Prerequisites	Corequisites
Machine Learning Foundations	4	BSCS2004	None	None
Business Data Management	4	BSMS2001	None	None

Machine Learning Techniques	4	BSCS2007	None	BSCS2004
Machine Learning Practice	4	BSCS2008	BSCS2004, BSCS2007	None
Machine Learning Practice - Project	2	BSCS2008P	None	BSCS2008
Tools in Data Science	3	BSSE2002	None	BSCS2008
Business Analytics (Option 1)	4	BSMS2002	BSMS2001	None
Business Data Management Project (Option 1)	2	BSMS2001P	None	BSMS2001
Intro to Deep Learning & GenAI (Option 2)	4	BSDA2001	None	BSCS2008
Deep Learning & GenAI Project (Option 2)	2	BSDA2001P	BSCS2007	BSCS2008, BSDA2001

Detailed Syllabus of Diploma Level Courses:

Courses for Diploma in Programming (Sr. No. 9 -16)

9. Database Management Systems-

A comprehensive introduction to databases, database management, and relevant topics like database security, integrity, concurrency, and data warehousing.

Course ID: BSCS2001

Course Credits: 4

Course Type: Programming

Pre-requisites: None

12 weeks of coursework, weekly online assignments, 2 in-person invigilated quizzes, 1 in-person invigilated end term exam. For details of standard course structure and assessments.

WEEK 1	Course Overview
WEEK 2	Relational Model and Basic SQL
WEEK 3	Intermediate and Advanced SQL
WEEK 4	Relational Query Languages and Database Design
WEEK 5	Functional Dependency and Normal Forms
WEEK 6	Functional Dependency and Normal Forms (cont.)
WEEK 7	Application Development
WEEK 8	Storage Management
WEEK 9	Indexing and Hashing
WEEK 10	Transactions
WEEK 11	Backup and Recovery
WEEK 12	Query Optimization and Conclusion

Prescribed Books

The following are the suggested books for the course:

Database Management Systems - Abraham Silberschatz, Henry F. Korth, S. Sudarshan

10. Programming, Data Structures and Algorithms using Python

A good foundation course to introduce basic concepts in the design and analysis of algorithms as well as standard data structures, using

Python as a base language for implementing these.

Course ID: BSCS2002

Course Credits: 4

Course Type: Programming

Pre-requisites: None

12 weeks of coursework, weekly online assignments, 2 in-person invigilated quizzes, 1 in-person invigilated end term exam. For details of standard course structure and assessments, visit Academics page.

WEEK 1	Python Refresher
WEEK 2	Complexity, Notations, Sorting and Searching Algorithms
WEEK 3	Arrays, Lists, Stacks, Queues, Hashing
WEEK 4	Graph Algorithms
WEEK 5	Graph Algorithms (Continued)
WEEK 6	Union-Find Data Structure, Priority Queue, Heap, Binary Search Tree
WEEK 7	Balanced Search Tree, Greedy Algorithms
WEEK 8	Divide and Conquer
WEEK 9	Dynamic Programming
WEEK 10	String or Pattern Matching Algorithms
WEEK 11	Network Flows, Linear Programming, Class of Algorithms
WEEK 12	Summary

11. Modern Application Development I

Building a modern application involves many different aspects: front end, recording transactions, storage, connecting to a remote server, using APIs etc. The courses Modern Application Development I and II go through all these aspects through a detailed and evolving case study, teaching the relevant programming skills as the course progresses.

Course ID: BSCS2003

Course Credits: 4

Course Type: Programming

Pre-requisites: None

Co-requisites: BSCS2001 - Database Management Systems

12 weeks of coursework, weekly online assignments, 2 in-person invigilated quizzes, 1 in-person invigilated end term exam. For details of standard course structure and assessments.

WEEK 1	Basic terminologies of Web
WEEK 2	Webpages written in HTML and CSS
WEEK 3	Presentation layer - View
WEEK 4	Models - Introduction to databases
WEEK 5	Controllers - Business logic
WEEK 6	APIs and REST APIs
WEEK 7	Backend Systems
WEEK 8	Application Frontend
WEEK 9	Application Security
WEEK 10	Testing of Web Applications
WEEK 11	HTML Evolution and Beyond HTML
WEEK 12	Application Deployment

12. Modern Application Development I – Project

"Modern Application Development - I project" is a comprehensive course designed to introduce learners to the fundamentals of web application development. The course covers essential technologies such as HTML, CSS, JavaScript, Flask, API implementation, and SQLite for data storage. Learners will gain practical knowledge in building secure, efficient, and deployable web applications while understanding key concepts in web development".

Course ID: BSCS2003P

Course Credits: 2

Course Type: Programming

Pre-requisites: None

Co-requisites: BSCS2003 - Modern Application Development I

Modern Application Development - I project" is a comprehensive course designed to introduce learners to the fundamentals of web application development. The course covers essential technologies such as HTML, CSS, JavaScript, Flask, API implementation, and SQLite for data storage. Learners will gain practical knowledge in building secure, efficient, and deployable web applications while understanding key concepts in web development.

13. Programming Concepts using Java

This course uses Java to provide an understanding of core ideas in object oriented programming, exception handling, event driven programming, concurrent programming and functional programming.

Course ID: BSCS2005

Course Credits: 4

Course Type: Programming

Pre-requisites: None

WEEK 1	**Basic Object Oriented Programming: Class Hierarchy**
WEEK 2	Basic Object Oriented Programming: Inheritance, Overriding
WEEK 3	Basic Object Oriented Programming: Polymorphism
WEEK 4	Basic Object Oriented Programming: Abstract Classes
WEEK 5	Collections. Iterators.
WEEK 6	Generics. Callbacks.
WEEK 7	Cloning. I/O serializations. Packages
WEEK 8	Cloning. I/O serializations. Packages (Continued)
WEEK 9	Exception handling
WEEK 10	Concurrent programming
WEEK 11	Concurrent programming (Continued)
WEEK 12	Concurrent programming (Continued)

14. Modern Application Development II

Building a modern application involves many different aspects: front end, recording transactions, storage, connecting to a remote server, using APIs etc. The courses Modern Application Development I and II go through all these aspects through a detailed and evolving case study, teaching the relevant programming skills as the course progresses.

Course ID: BSCS2006

Course Credits: 4

Course Type: Programming

Pre-requisites: BSCS2003 - Modern Application Development I

12 weeks of coursework, weekly online assignments, 2 in-person invigilated quizzes, 1 in-person invigilated end term exam. For details of standard course structure and assessments-

WEEK 1	Basics of JavaScript
WEEK 2	Advanced JavaScript
WEEK 3	Introduction to Web Frontend
WEEK 4	Introduction to VueJS
WEEK 5	Vue with APIs
WEEK 6	Advanced Vuejs
WEEK 7	Advanced State Management
WEEK 8	Authentication and Designing APIs
WEEK 9	Asynchronous Jobs
WEEK 10	Inter-Service Messaging and Webhooks
WEEK 11	Performance
WEEK 12	Project

15. Modern Application Development II - Project

Course ID: BSCS2006P

Course Credits: 2

Course Type: Programming

Pre-requisites: BSCS2003P - Modern Application Development I - Project

Co-requisites: BSCS2006 - Modern Application Development II

"Modern Application Development - II project" is an advanced course designed to build upon the foundations of web application development introduced in "Modern Application Development - I." The course focuses on advanced frontend technologies, such as Vue.js, to create interactive and sophisticated user interfaces. It also covers additional concepts like token-based authentication, asynchronous task execution using Celery and Redis, caching, and security considerations. The course covers some important concepts of web like Token-based Authentication, Asynchronous Task Execution with Celery and Redis, Caching Techniques. The course also talks about Security and Privacy Considerations and Cross-Origin Resource Sharing (CORS).

16. System Commands

Course ID: BSSE2001

Course Credits: 3

Course Type: Programming

Pre-requisites: None

8 weeks of coursework, weekly online assignments, 2 in-person invigilated quizzes, 1 in-person invigilated end term exam. For details of standard course structure and assessments, visit Academics page.

WEEK 1 Introduction to GNU/Linux OS. Setting up and running Linux environment. The command line environment. Knowing hardware of your machine Information - commands such as hwinfo, lshw, df, free etc. Diagnostics - commands to fetch hardware information such as battery state, memory modules etc., Knowing the OS and software of your machine Commands to get details about operating system, versions etc. Packages - installed / available Input / output redirection.

WEEK 2 Introduction to packages and repositories. Using 'apt' commands to manage packages. File types and related commands. Understanding file permissions and access modes. Managing file permissions through symbolic and numeric mode. Concept of environment variables. Important environment variables such as $HOME, $USER and $PATH

WEEK 3 Managing shell variables. Prompt strings. Symbolic links and hard links, brief introduction to inode numbers. Exploring the root file system and related commands. Using shell shortcuts with commands. Slicing output. Managing programs currently running on the machine. Shell access to a local / remote machine.

WEEK 4 Redirection to script, variable and for logging purpose. Using pipes. Introduction to regex; using regex patterns and grep. Using grep to extract useful information from files. find command and its uses - patterns to pick specific files in a folder, using exec Command line editors (nano, vi, emacs - syntax highlighting & prompting, configuring options) Writing and running simple Bash scripts.

WEEK 5 How are shell scripts interpreted? Using variables in scripts. Passing command line arguments to scripts, to create your own commands. More shell programming. Writing conditional statements using if / else / fi. Introduction to loops. Using functions Configuring startup / periodic / recurring tasks

WEEK 6 Text processing using AWK language. Using awk to run statics on a data file. Using regex within awk. Awk as a programming language Introduction to 'sed' – another text processing utility. Line by line processing to replace a regex pattern with a string. Use of place holders for matching regex patterns for use in replacing strings.

WEEK 7 Introduction to make utility; concept of target and dependency; actions performed by make; conditional compilation; passing shell variables to make; File packaging utilities such as compress, tar, zip, gzip, bzip2, xz. Networking concepts Introduction to IP addresses. Concept of localhost. Intranet and public addresses. Concept of ports and services that run on these ports. Concept of DNS and Domain names. Network diagnostics using tools and commands. Scripting a tool for analysis of logs.

WEEK 8 Introduction to RAID for handling hardware failure. Introduction to version control Git as a version control system: Overview of Git workflow Branches, repositories, forks, etc. Creating and merging pull requests. Personal access tokens. Managing changes as local and remote. Working demo of a public repository using Git Approval workflow How a team collaborates on the private repository in an organization. Managing pull requests for the owner, raising issues and resolving them.

Courses for Diploma in Programming (Sr. No. 17 - 26)

17. Machine Learning Foundations

This course lays the groundwork for the upcoming ML courses by covering various fundamentals that do not necessarily fall under Machine Learning but are quite necessary for a comprehensive understanding of Machine Learning.

Course ID: BSCS2004

Course Credits: 4

Course Type: Data Science

Pre-requisites: None

12 weeks of coursework, weekly online assignments, 2 in-person invigilated quizzes, 1 in-person invigilated end term exam. For details of standard course structure and assessments.

WEEK 1	**Introduction to machine learning**
WEEK 2	Calculus
WEEK 3	Linear Algebra - Least Squares Regression
WEEK 4	Linear Algebra - Eigenvalues and eigenvectors
WEEK 5	Linear Algebra - Symmetric matrices
WEEK 6	Linear Algebra - Singular value decomposition, Principal Component Analysis in Image Processing
WEEK 7	Unconstrained Optimisation
WEEK 8	Convex sets, functions, and optimisation problems
WEEK 9	Constrained Optimisation and Lagrange Multipliers. Logistic regression as an optimization problem
WEEK 10	Examples of probabilistic models in machine learning problems
WEEK 11	Exponential Family of distributions
WEEK 12	Parameter estimation. Expectation Maximization.

18. Business Data Management

A significant source of data sets and problems for data scientists will come from the business domain. This course provides a basic understanding of how businesses are organised and run from a data perspective.

Course ID: BSMS2001

Course Credits: 4

Course Type: Data Science

Pre-requisites: None

12 weeks of coursework, weekly online assignments, 2 in-person invig-

ilated quizzes, 1 in-person invigilated end term exam. For details of standard course structure and assessments, visit Academics page.

WEEK 1 Consumption and demand: Micro & Macro economics: the role of data, production, consumption and exchange, consumption baskets, sources of consumer survey data

WEEK 2 Micro-economic concepts: Utility: cardinal vs ordinal, indifference curves. Demand and supply curves, changes in demand and elasticity. production cost, cost curves. Make vs buy decisions, production quantity decisions

WEEK 3 Firm level strategies and performance data: Objectives and types of pricing strategies, analysis of firm performance - key ratios. Analysis examples: Ultratech, Page Industries, Nestle, TCS

WEEK 4 Analysing industry level data: Industry definition and classification codes, IIP and PMI, industry market structure and concentration indices, competitive positioning in an industry - Porter's five forces. Analysis examples: Cement industry, Textile industry, FMCG industry, IT industry

WEEK 5 Case study 1 - Fabmart (E-Commerce): Introduction to E-Commerce, Fabmart case introduction, explanation of data set & questions to be answered, revenue pareto, volume pareto, scatter plot of sales and revenue, revenue trend

WEEK 6 Fabmart case continued: Sales analysis, organisation of distribution centre, analysis of sales trends, average days of inventory, ledger, avoiding stockouts

WEEK 7 Case study 2 - Ace Gears (Manufacturing): Introduction to the manufacturing sector, context of the automotive industry during the years 2019-2021, explanation of data set containing monthly information on sales, production, inventory and costing. Revenue trend analysis, portfolio management

WEEK 8 Ace Gears case study continued: Regional sales analysis, sales agent planning, production scheduling, scrap analysis, unit level profitability analysis, raw material re-ordering and safety stock

WEEK 9 Case study 3 - Tech Enterprises (IT): Introduction to HR as a function, Introduction to the Tech Enterprises, internal sourcing, ranking of internal candidates, job description, sourcing channels and their analysis, recruitment process and onboarding

WEEK 10 Case study 4 - PayBuddy (Fin Tech): Introduction to Finance Industry and Fintech, payment processing and money flow, new credit product introduction, nudge economics, payment transaction and customer data set, identifying rules to target the appropriate customers

WEEK 11 Paybuddy case continued: Introduction to A/B testing, analysis of the A/B testing data, credit risk evaluation, risk-return tradeoffs

WEEK 12 Discussion on student acquired data sets. Wrap up (summary) of the case studies, course project work

19. Machine Learning Techniques

To introduce the main methods and models used in machine learning problems of regression, classification and clustering. To study the properties of these models and methods and learn about their suitability for different problems.

Course ID: BSCS2007

Course Credits: 4

Course Type: Data Science

Pre-requisites: None

Co-requisites: BSCS2004 - Machine Learning Foundations

12 weeks of coursework, weekly online assignments, 2 in-person invigilated quizzes, 1 in-person invigilated end term exam. For details of standard course structure and assessments.

WEEK 1	**Introduction; Unsupervised Learning - Representation learning - PCA**
WEEK 2	Unsupervised Learning - Representation learning - Kernel PCA
WEEK 3	Unsupervised Learning - Clustering - K-means/Kernel K-means
WEEK 4	Unsupervised Learning - Estimation - Recap of MLE + Bayesian estimation, Gaussian Mixture Model - EM algorithm.
WEEK 5	Supervised Learning - Regression - Least Squares; Bayesian view
WEEK 6	Supervised Learning - Regression - Ridge/LASSO
WEEK 7	Supervised Learning - Classification - K-NN, Decision tree
WEEK 8	Supervised Learning - Classification - Generative Models - Naive Bayes
WEEK 9	Discriminative Models - Perceptron; Logistic Regression
WEEK 10	Support Vector Machines
WEEK 11	Ensemble methods - Bagging and Boosting (Adaboost)
WEEK 12	Artificial Neural networks: Multiclass classification.

Prescribed Books-

The following are the suggested books for the course:

Pattern Classification by David G. Stork, Peter E. Hart, and Richard O. Duda

Pattern Recognition and Machine Learning by Christopher M. Bishop

The Elements of Statistical Learning: Data Mining, Inference, and Prediction by Trevor Hastie, Robert Tibshirani, and Jerome Friedman

20. Machine Learning Practice

This companion course to the ML Theory course introduces the student to scikit-learn, a popular Python machine learning module, to provide hands-on problem solving experience for all the methods and models learnt in the Theory course.

Course ID: BSCS2008

Course Credits: 4

Course Type: Data Science

Pre-requisites: BSCS2004 - Machine Learning Foundations

BSCS2007 - Machine Learning Techniques

12 weeks of coursework, weekly online assignments, 2 in-person invigilated quizzes, 1 in-person invigilated end term exam. For details of standard course structure and assessments.

WEEK 1	**End-to-end machine learning project on scikit-learn**
WEEK 2	Graph Theory(VOL 3)
WEEK 3	Regression on scikit-learn - Linear regression Gradient descent - batch and stochastic.
WEEK 4	Polynomial regression, Regularized models
WEEK 5	Logistic regression
WEEK 6	Classification on scikit-learn - Binary classifier
WEEK 7	Classification on scikit-learn - Multiclass classifier
WEEK 8	Support Vector Machines using scikit-learn
WEEK 9	Decision Trees, Ensemble Learning and Random Forests
WEEK 10	Decision Trees, Ensemble Learning and Random Forests (Continued)
WEEK 11	Neural networks models in scikit-learn
WEEK 12	Unsupervised learning

21. Machine Learning Practice – Project

The purpose of a machine learning project is to apply ML models learnt in ML courses, on real world data and to create an effective predictive model. Discover patterns in your data and then make predictions based on often complex findings to answer business questions, detect and analyze trends and help solve problems.

Course ID: BSCS2008P

Course Credits: 2

Course Type: Data Science

Pre-requisites: None

Co-requisites: BSCS2008 - Machine Learning Practice

Project Course Overview-

The MLP Project is conducted on Kaggle platform.

Kaggle is a well recognized platform among data science professionals that will familiarize you with many latest problems being currently solved by organizations across the world.

Good place to practice your ML skills and knowledge. Looks good on your resume, especially if you have won medals and badges.

The project runs for 12 to 15 weeks. The project is divided into several milestones. These milestones divide the ML cycle into smaller and manageable steps. These milestones guide you to make steady progress.

22. Tools in Data Science

This practical course will teach students to use popular tools for sourcing data, transforming it, analyzing it, communicating these as visual stories, and deploying them in production.

Pre-requisites: Python, HTML, JavaScript, Excel, data science basics.

Course ID: BSSE2002

Course Credits: 3

Course Type: Data Science

Pre-requisites: None

Co-requisites: BSCS2008 - Machine Learning Practice

9 modules over 12 weeks of coursework, online assignments for each module, 1 remote online exam, 2 take home projects, 1 in-person end term exam. For details of standard course structure and assessments.

MODULE 1	**Development Tools and concepts to build models and apps.**
MODULE 2	Deployment Tools and concepts to publish what you built.
MODULE 3	Large Language Models that make your work easier and your apps smarter.
MODULE 4	Data Sourcing to get data from the web, files, and databases.
MODULE 5	Data Preparation to clean up and convert the inputs to the right format.
MODULE 6	Data Analysis to find surprising insights in the data.
MODULE 7	Data Visualization to communicate those insights as visual stories.

23. Business Data Management – Project (Option 1)

BDM Capstone Project is an independent research project where the student is expected to reach out to a business firm (either from organized sectors viz., well established businesses in manufacturing, IT, automobile sectors etc. which has excellent systems in place to handle and manage data or from an unorganized sectors like Kirana stores, vegetable vendors etc. which do not maintain proper records), identify the issues or the problem(s) they face, collect primary data pertaining to it, clean the data, analyze it, and provide novel / valuable insights to the decision maker(s).

Course ID: BSMS2001P

Course Credits: 2

Course Type: Data Science

Pre-requisites: None

Co-requisites: BSMS2001 - Business Data Management

Project Course Overview

In the proposal stage, the student is expected to provide information about the organization and its background, problem definition/ statement, the background of the problem, the problem-solving approach they wish to use with justification, the expected timelines for project completion and the probable outcomes of the project.

In the midterm submission, students must provide a short video interaction clip with the business owner/ manager, tangible evidence like pictures, field notes etc., provide information on the meta data and descriptive statistics, conduct preliminary analysis, provide an interpretation of the results and findings.

The final report submission is a complete report and involves in-depth explanation of the entire process from start to finish.

The viva voce is conducted following acceptance of the final submission and for this, the student needs to make a presentation not exceeding 10 slides.

24. Business Analytics (Option 1)

The problems faced by decision makers in today's business environments are extremely complex. Hence, the task of making good decisions is not easy. The answer is in building quantitative models, and this course is designed to help you understand the fundamentals of this critical, foundational, business skill. The business application of statistical methods is the core focus of this course. In that sense, the course builds on the core course in the first year of the program. That basic course focused on the preliminaries of the area. This course highlights a business application and then demonstrates an application of a statistical technique to solve that scenario and arrive at the best decisions and insights.

Course ID: BSMS2002

Course Credits: 4

Course Type: Data Science

Pre-requisites: BSMS2001 - Business Data Management

8 weeks of coursework, weekly online assignments, 2 in-person invigilated quizzes, 1 in-person invigilated end term exam. For details of standard course structure and assessments.

WEEK 1	**Data dashboarding Insights from data summary**
WEEK 2	Can summarizing the data provide insights?
WEEK 3	Do people in different cities prefer different brand?
WEEK 4	Predicting the stock returns – Regression basics
WEEK 5	How do you pay a professor? – Regression diagnostics - Path variables
WEEK 6	Can I cure cancer? – Logistic Regression - Connection with classification problem
WEEK 7	What is the impact of repeatedly watching the same ad? Repeated measures ANOVA
WEEK 8	When the data has a time axis: Time series modeling

Prescribed Books-

The following are the suggested books for the course:

SF Robert E Stine and Dean Foster, Statistics for Business: Decision Making and Analysis, Pearson.

NCT Paul Newbold, William L. Carlson and Betty Thorne, Statistics for Business and Economics, Sixth Edition, Pearson.

ASW David R. Anderson, Dennis J. Sweeney and Thomas A. Williams, Statistics for Business and Economics, Ninth Edition, Cengage Learning.

Keller Gerald Keller, Managerial Statistics, 9 Edition, Cengage Learning.

LR Richard Levin and David Rubin, Statistics for Management, Seventh Edition, Pearson.

25. Introduction to DL and GenAI (Option 2)

This course aims to provide a comprehensive introduction to the foundational and practical aspects of Deep Learning and Generative AI. Through a balanced blend of theoretical concepts and hands-on experience, students will learn to build, train, and evaluate artificial neural networks for a variety of tasks in computer vision and natural language processing. The course covers key architectures such as Convolutional Neural Networks (CNNs) for image data, Recurrent Neural Networks (RNNs) and LSTMs for sequential data, and extends into the realm of generative models including Autoencoders, Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs) and Large Language Models (LLMs). By the end of the course, learners will gain the skills to implement core deep learning models and apply generative AI techniques to solve practical problems.

Course ID: BSDA2001

Course Credits: 4

Course Type: Data Science

Pre-requisites: None

Co-requisites: BSCS2008 - Machine Learning Practice

WEEK 1 Artificial Neural Networks - Theory Introduction to Deep Learning, Artificial neurons, neural networks, layers, Activation functions and loss metrics

WEEK 2 Artificial Neural Networks - Practice Hands-on: Build simple neural networks using Tensor Flow/Keras, Experimentation with activation functions and optimization methods

WEEK 3 Modeling Vision — CNN - Theory Introduction to Convolutional Neural Networks (CNNs), CNN architecture basics, convolution and pooling layers

WEEK 4 Modeling Vision — CNN - Practice Hands-on: CNN-based image classification (e.g., MNIST, CIFAR-10)

WEEK 5 Modeling Sequential Data - Theory Sequence models, Recurrent Neural Networks (RNNs), LSTMs

WEEK 6 Modeling Sequential Data - Practice Hands-on: Sentiment analysis with LSTM, Text generation with simple RNN/LSTM models, Time series prediction tasks

WEEK 7 Generative AI for vision — Variational AutoEncoders and GANs - Theory Introduction to Generative AI, Autoencoders and Variational Autoencoders (VAEs) basics, GANs

WEEK 8 Generative AI for vision—Diffusion Models Diffusion Probabilistic models, training and inference

WEEK 9 Generative AI for vision— Practice GANs and Pretrained Diffusion Models to generate Images

WEEK 10 Large Language Models — Transformer Architecture, Word Embeddings, Tokenization, Attention

WEEK 11 Large Language Models - encoder, decoder, encoder decoder models, BERT like models for NLP tasks, Decoders for Text Generation, Machine Translation, Fine-Tuning LLMs (PEFT, LoRA)

WEEK 12 Large Language Models - Practice Prompting Techniques, Prompt Fine-tuning and other methods

26. Introduction to DL and GenAI Project

Course ID: BSDA2001P

Course Credits: 2

Course Type: Data Science

Pre-requisites: BSCS2007 - Machine Learning Techniques

Co-requisites: BSCS2008 - Machine Learning Practice

BSDA2001 - Introduction to DL and GenAI

What you'll learn -

Hands-on experience with deep learning frameworks like PyTorch, Huggingface.

Working with large language models (LLMs) and generative AI models

(text, image, etc.)

Applying transformers, diffusion models, and fine-tuning techniques

Enhancing your profile with a Kaggle competition medal and an industry-relevant AI project

Preparing for technical viva conducted by an industry expert to simulate real interview conditions.

3 Degree Level

BSc Degree Level (Programming & Data Science)

Requirements: Completion of Foundation (8 courses) + Diploma Level (12 courses + 4 projects).

Credits: 28 (Total 114 credits) | Duration: 1–3 years | Workload: 15 hrs/week

Outcome: Eligible learners may proceed to BS Degree Level or exit with BSc Degree.

BS Degree Level (Data Science and Applications)

Requirements: Completion of 114 credits and BSc Degree Level.

Credits: 28 (Total 142 credits) | Duration: 1–3 years | Workload: 15 hrs/week

Outcome: Exit with BS Degree or proceed to PG Diploma (CGPA $\geq$ 8.0).

Core Courses

Pair 1

- Software Engineering
- Software Testing

Pair 2

- AI: Search Methods for Problem Solving
- Deep Learning

Elective Courses: 38 Courses

Detailed Syllabus of Degree Level Courses: (Sr. No. 27 Onwards...)

Pair 1

27. Software Engineering

To prepare students to develop the essential skills required to become effective software engineers by introducing them to fundamental concepts in developing software, and essential practices employed by software developers, such as requirement gathering, creating software conceptual designs, software comprehension, debugging, testing and deployment.

Course ID: BSCS3001

Course Credits: 4

Course Type: Core Option I

Pre-requisites: None

12 weeks of coursework, weekly online assignments, 2 in-person invigilated quizzes, 1 in-person invigilated end term exam. For details of standard course structure and assessments.

WEEK 1	**Deconstructing the software development process**
WEEK 2	Identify different types of software requirements (functional, non-functional)
WEEK 3	Software Conceptual Design
WEEK 4	Software Usability
WEEK 5	Software Design - Modeling and Architecture
WEEK 6	Software Design - Quality and Evaluation
WEEK 7	Software Development - Program Comprehension
WEEK 8	Software Development - Program Debugging
WEEK 9	Software - Code Reviewing and Documentation
WEEK 10	Software Testing
WEEK 11	Software Deployment and Monitoring
WEEK 12	Conclusion and other Aspects: Communication, Productivity and Organizations

Prescribed Books

The following are the suggested books for the course:

Software Engineering: A Precise Approach – Dr. Pankaj Jalote

Cooperative Software Development – Dr. Amy Ko

28. Software Testing

To prepare the students to understand the phases of testing based on requirements for a project, to apply the concepts taught in the course to formulate test requirements precisely, to design and execute test cases as a part of a standard software development IDE, and to apply specially designed test case design techniques for specific application domains.

Course ID: BSCS3002

Course Credits: 4

Course Type: Core Option I

Pre-requisites: None

12 weeks of coursework, weekly online assignments, 2 in-person invigilated quizzes, 1 in-person invigilated end term exam. For details of standard course structure and assessments, visit Academics page.

WEEK 1 Software Testing: Motivation, Software Development Life Cycle, Terminologies and Processes, Software Test Automation: JUnit as an example

WEEK 2 Basics of Graphs, Fundamental Graph Algorithms, Elementary Graph Algorithms, Structural Graph Coverage Criteria, Algorithms: Structural Graph Coverage Criteria

WEEK 3 Graph Coverage Criteria: Applied to Test Code, Data Flow in Graphs,, Data Flow Testing Example, Unit Testing Based on Graphs: Summary

WEEK 4 Software Design and Integration Testing, Design Integration Testing and Graph Coverage, Specification Testing and Graph Coverage, Graph Coverage and Finite State Machines (FSM), Testing Source Code: Classical Coverage Criteria

WEEK 5 Logic: Basics needed for Software Testing, Coverage Criteria, Logic Coverage Criteria: Making clauses determine predicate, Applied to test code

WEEK 6 Logic: Coverage Example, Coverage Specification, Coverage FSM, Coverage Summary, SMT - Solvers

WEEK 7 Symbolic Testing, Concolic Execution, Example and Summary Symbolic Execution

WEEK 8 Requirements, Functional Testing, ISP, ISP Example

WEEK 9 Regular Expense CFGs, Mutation Testing, Mutation Operators Source Code, Mutation Testing Vs Other Criteria, Mutation Testing For Integration And Tools

WEEK 10 Basic Object Oriented (OO) Integration Concepts, Mutation Operators OO Integration, Mutation Operators OO Integration, OO Faults, Coupling Criteria

WEEK 11 Web Apps Intro, Client Side Testing, Server Side Testing

WEEK 12 Regression Testing, Software Quality Metrics, Non Functional Testing, TDD, Course Summary

Prescribed Books

The following are the suggested books for the course:

Paul Ammann and Jeff Offutt, Introduction to Software Testing, Cambridge University Press, 2008.

Glenford J. Myers, The Art of Software Testing, Second edition, 2008.

Paul C. Jorgensen, Software Testing: A Craftsman's Approach, Fourth edition, CRC Press, 2014.

Lisa Crispin and Janet Gregory, Agile Testing: A Practical Guide for Testers and Agile Teams, Addison-Wesley, 2009.

Appropriate research papers on testing techniques, information regarding testing tools, as applicable.

Pair 2:

29. AI: Search Methods for Problem Solving

We look at how an intelligent agent solves new problems. Starting with blind search we quickly move on to heuristic search, and look at several variations designed to combat the combinatorial explosion that search has to face. We study how board games like Chess and Go are played; how search facilitates logical reasoning; and approaches to domain independent planning of actions to achieve a goal. We end with looking at an alternative formulation that combines search and reasoning as constraint processing.

Course ID: BSCS3003

Course Credits: 4

Course Type: Core Option II

Pre-requisites: None

12 weeks of coursework, weekly online assignments, 2 in-person invig-

ilated quizzes, 1 in-person invigilated end term exam. For details of standard course structure and assessments, visit Academics page.

WEEK 1 Introduction and philosophy. The Turing Test. The Winograd Schema Challenge. Placing search in the landscape of AI.

WEEK 2 Search spaces. Examples. State space search. Depth First, Breadth First, Iterative Deepening. Analysis.

WEEK 3 Heuristic search. Heuristic functions. Solution space search. Escaping local optima. Stochastic local search.

WEEK 4 Population based methods. Genetic Algorithms, emergent systems, Ant Colony Optimization.

WEEK 5 Finding optimal paths. Algorithm A*. Admissibility of A*.

WEEK 6 The monotone condition. Space saving versions of A*. Sequence alignment.

WEEK 7 Game playing. Board games. Algorithms Minimax, Alpha-Beta, and SSS*.

WEEK 8 Automated domain independent planning. Goal Stack Planning, Partial Order Planning.

WEEK 9 Problem decomposition with goal trees. Algorithm AO*.

WEEK 10 Pattern directed inference systems. Forward chaining inference engine. The Rete algorithm.

WEEK 11 Constraint processing. Algorithm Backtracking. Arc consistency. Combining search and reasoning. Waltz algorithm. Model based diagnosis.

Prescribed Books

The following are the suggested books for the course:

Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education (India), 2013. (Chapters 1–8, some parts from Chapters 9 and 10)

Reference

John Haugeland, Artificial Intelligence: The Very Idea, A Bradford Book, The MIT Press, 1985.

Pamela McCorduck, Machines Who Think: A Personal Inquiry into the History and Prospects of Artificial Intelligence, A K Peters/CRC Press; 2nd edition, 2004.

Eugene Charniak and Drew McDermott, Introduction to Artificial Intelligence, Addison-Wesley Publ., 1985.

Zbigniew Michalewicz and David B. Fogel. How to Solve It: Modern Heuristics. Springer; 2nd edition, 2004.

Judea Pearl. Heuristics: Intelligent Search Strategies for Computer Problem Solving, Addison-Wesley, 1984.

Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill, 1991.

Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd Edition, Prentice Hall, 2009.

Patrick Henry Winston. Artificial Intelligence, Addison-Wesley, 1992.

Stefan Edelkamp and Stefan Schroedl. Heuristic Search: Theory and Applications, Morgan Kaufmann, 2011.

30. Deep Learning

To study the basics of Neural Networks and their various variants such as the Convolutional Neural Networks and Recurrent Neural Networks, to study the different ways in which they can be used to solve problems in various domains such as Computer Vision, Speech and NLP.

Course ID: BSCS3004

Course Credits: 4

Course Type: Core Option II

Pre-requisites: None

12 weeks of coursework, weekly online assignments, 2 in-person invigilated quizzes, 1 in-person invigilated end term exam. For details of

standard course structure and assessments, visit Academics page.

WEEK 1 History of Deep Learning, McCulloch Pitts Neuron, Thresholding Logic, Perceptron Learning Algorithm and Convergence

WEEK 2 Multilayer Perceptrons (MLPs), Representation Power of MLPs, Sigmoid Neurons, Gradient Descent

WEEK 3 Feedforward Neural Networks, Representation Power of Feedforward Neural Networks, Backpropagation

WEEK 4 Gradient Descent(GD), Momentum Based GD, Nesterov Accelerated GD, Stochastic GD, Adagrad, AdaDelta, RMSProp, Adam, AdaMax, NAdam, learning rate schedulers

WEEK 5 Autoencoders and relation to PCA, Regularization in autoencoders, Denoising autoencoders, Sparse autoencoders, Contractive autoencoders

WEEK 6 Bias Variance Tradeoff, L2 regularization, Early stopping, Dataset augmentation, Parameter sharing and tying, Injecting noise at input, Ensemble methods, Dropout

WEEK 7 Greedy Layer Wise Pre-training, Better activation functions, Better weight initialization methods, Batch Normalization

WEEK 8 Learning Vectorial Representations Of Words, Convolutional Neural Networks, LeNet, AlexNet, ZF-Net, VGGNet, GoogLeNet, ResNet

WEEK 9 Visualizing Convolutional Neural Networks, Guided Backpropagation, Deep Dream, Deep Art, Fooling Convolutional Neural Networks

WEEK 10 Recurrent Neural Networks, Backpropagation Through Time (BPTT), Vanishing and Exploding Gradients, Truncated BPTT

WEEK 11 Gated Recurrent Units (GRUs), Long Short Term Memory (LSTM) Cells, Solving the vanishing gradient problem with LSTM

WEEK 12 Encoder Decoder Models, Attention Mechanism, Attention over images, Hierarchical Attention, Transformers.

Prescribed Books

The following are the suggested books for the course:

Ian Goodfellow and Yoshua Bengio and Aaron Courville. Deep Learning. An MIT Press book. 2016.

Charu C. Aggarwal. Neural Networks and Deep Learning: A Textbook. Springer. 2019.

Elective Courses

Here is the list of elective courses offered in the program. In the BSc and BS level, a maximum of 8 credits can be transferred from NPTEL and there is the option to do an apprenticeship and transfer up to a maximum of 12 credits in the BS level.

(Note: List of elective courses may change each term depending on availability.)

Course Name	Code	Credits
1. Software Engineering CORE COURSE	BSCS3001	4
2. Software Testing CORE COURSE	BSCS3002	4
3. AI: Search Methods for Problem Solving CORE COURSE	BSCS3003	4
4. Deep Learning CORE COURSE	BSCS3004	4
5. Strategies for Professional Growth MANDATORY COURSE	BSGN3001	4
6. Algorithmic Thinking in Bioinformatics	BSBT4001	4
7. Big Data and Biological Networks	BSBT4002	4
8. Data Visualization Design	BSCS4001	4
9. Speech Technology	BSEE4001	4
10. Design Thinking for Data-Driven App Development	BSMS4002	4
11. Industry 4.0	BSMS4001	4
12. Market Research	BSMS3002	4
13. Privacy & Security in Online Social Media	BSCS4003	4
14. Introduction to Big Data	BSDA5001	4
15. Financial Forensics	BSMS4003	4

16. Linear Statistical Models	BSMA3012	4
17. Advanced Algorithms	BSCS4021	4
18. Statistical Computing	BSMA3014	4
19. Computer Systems Design	BSCS3031	4
20. Programming in C	BSCS3005	4
21. Mathematical Thinking	BSMA2001	4
22. Large Language Models	BSDA5004	4
23. Introduction to Natural Language Processing (i-NLP)	BSDA5005	4
24. Deep Learning for Computer Vision	BSDA5006	4
25. Managerial Economics	BSMS3033	4
26. Game Theory and Strategy	BSMS4023	4
27. Corporate Finance	BSMS3034	4
28. Deep Learning Practice	BSDA5013	4
29. Operating Systems	BSCS4022	4
30. Mathematical Foundations of Generative AI	BSDA5002	4
31. Algorithms for Data Science (ADS)	BSDA5003	4
32. Machine Learning Operations (MLOps)	BSDA5014	4
33. Data Science and AI Lab	BSDA4001	4
34. App Dev Lab	BSCS4010	4
35. Computer Networks	BSCS4024	4
36. Theory of Computation	BSCS3021	4
37. Reinforcement Learning	BSDA5007	4
38. Sequential Decision Making	BSDA6004	4

Detailed Syllabuses of the Elective Courses are available on the link given below-

https://study.iitm.ac.in/ds/academics.html

....End..